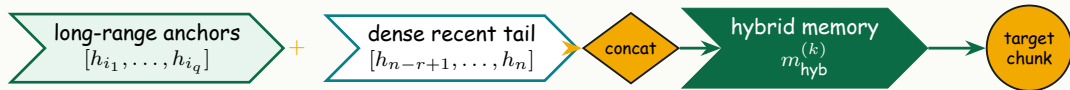
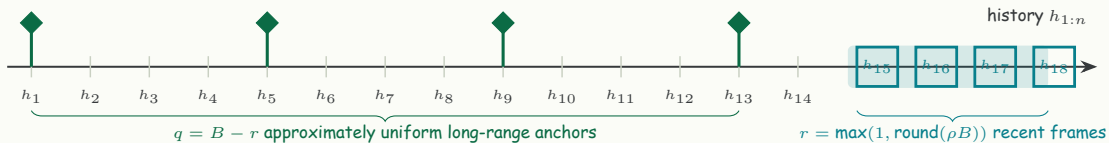


Hybrid-Sparse Memory under a Fixed Frame Budget

Emerald pins preserve long-range coverage; the teal band keeps a contiguous recent tail for local motion.



$$m_{\text{hyb}}^{(k)} = [h_{i_1}, \dots, h_{i_q}, h_{n-r+1}, \dots, h_n], \quad q + r = B$$